MILITARY SPECIFICATION SHEET

HEAT EXCHANGER, NBC SYSTEM
HEAT EXCHANGER P/N 12336771

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the Heat Exchanger 12336771 described herein shall consist of this document and the latest issue of specification MIL-H-62411(AT).

3. REQUIREMENTS

3.4 Design and construction. The heat exchanger shall conform to drawing hierarchy of which Drawing 12336771 is the top drawing.

3.4.1 Weight. The weight of the heat exchanger shall not exceed 13.2 pounds (lb) (see 4.6.3).

3.5 Performance.

3.5.1 Leakage.

3.5.1.1 External leakage. The pneumatic pressure applied to the hot air passages of the heat exchanger unit shall be 62 pounds per square inch gage (psig) (see 4.6.4.1).

3.5.1.2 Internal leakage. The pressure applied to the hot air passages shall be 62 psig. The pressure loss in the first 20 second shall not exceed 25 psig (see 4.6.4.2).

3.5.2 Proof pressure. The proof pressure shall be 93 (+3, -0) psig (see 4.6.5).

FSC 4130

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TOOL BOX Manufacturer's standard TOOL BOX shall be

3.5.3 Burst pressure. The burst pressure shall be 117 psig at 360 degrees Fahrenheit (°F) (see 4.6.6).

3.5.4 Heat transfer. Heat transfer performance shall be not less than that shown in figure 1 (see 4.6.7).

3.5.5 Air pressure loss. The pressure loss shall not exceed that shown in figure 2 (see 4.6.8).

4.6.4 Leakage.

4.6.4.1 External leakage. A pneumatic pressure of 62 to 64 psig shall be applied to the hot air passages.

4.6.4.2 Internal leakage. A penumatic pressure of 62 to 64 psig shall be applied to the hot air passages.

4.6.5 Proof pressure. The test pressure shall be 93 to 96 psig.

4.6.6 Burst pressure. The oven temperature shall be 360°F. The test pressure shall be 117 to 121 psig.

4.6.7 Heat transfer. To determine conformance to 3.5.4, the heat exchanger shall operate at each combination of the steady state conditions specified in table VI. The measured hot air outlet temperature shall be equal to or greater than that specified in table VI.

TABLE VI. Heat transfer.

Initial air temperature		Air flow		Hot air Outlet temperature °F ± 10°F
Hot	Cold	Hot	Cold	
210		10	15	174
210			30	200
210			52.2	206
210		14.7	15	150
210			30	190
210			52.2	202
210		20	15	147
210			30	180
210			52.2	202

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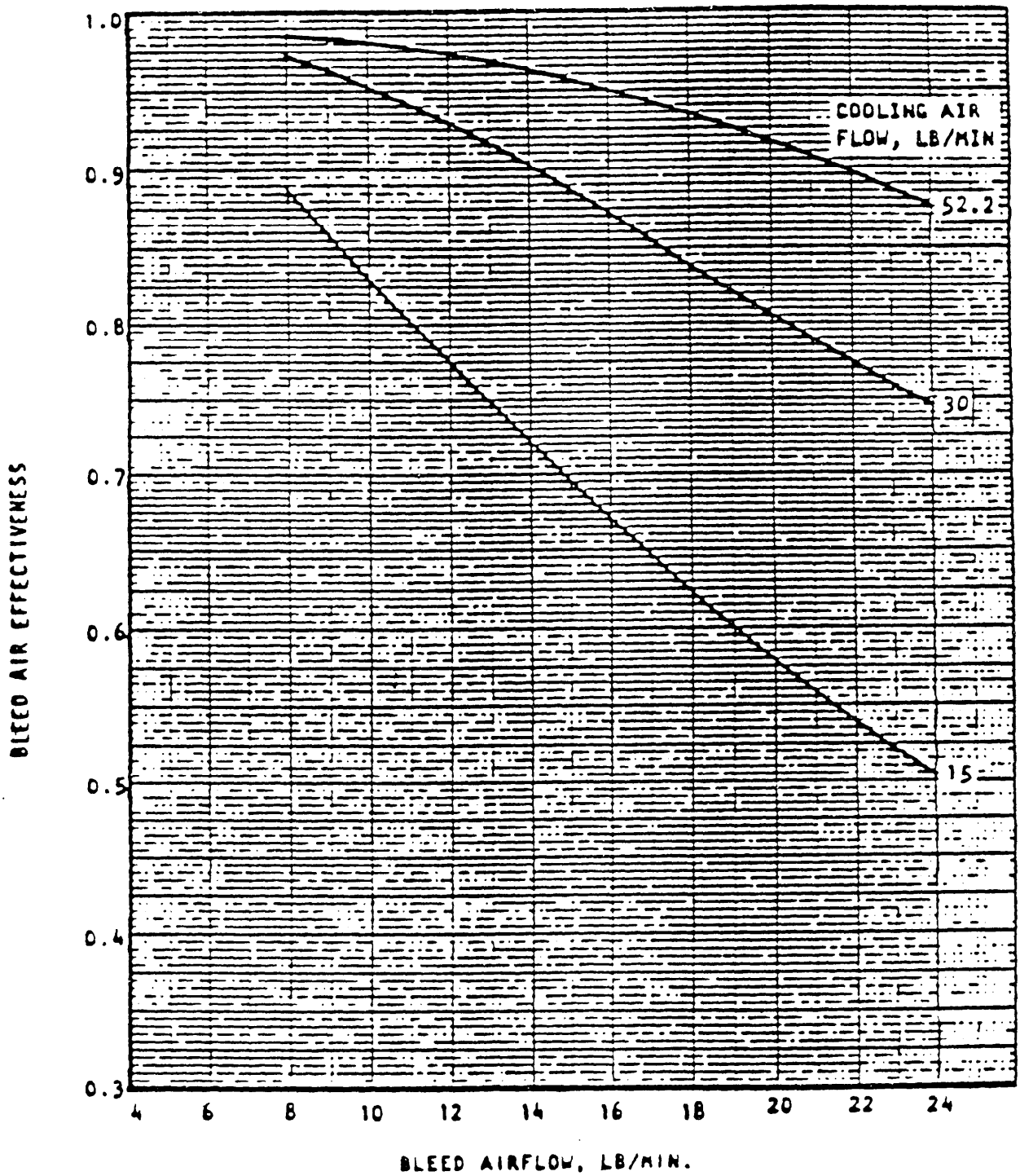


FIGURE 1. Required heat exchanger performance.

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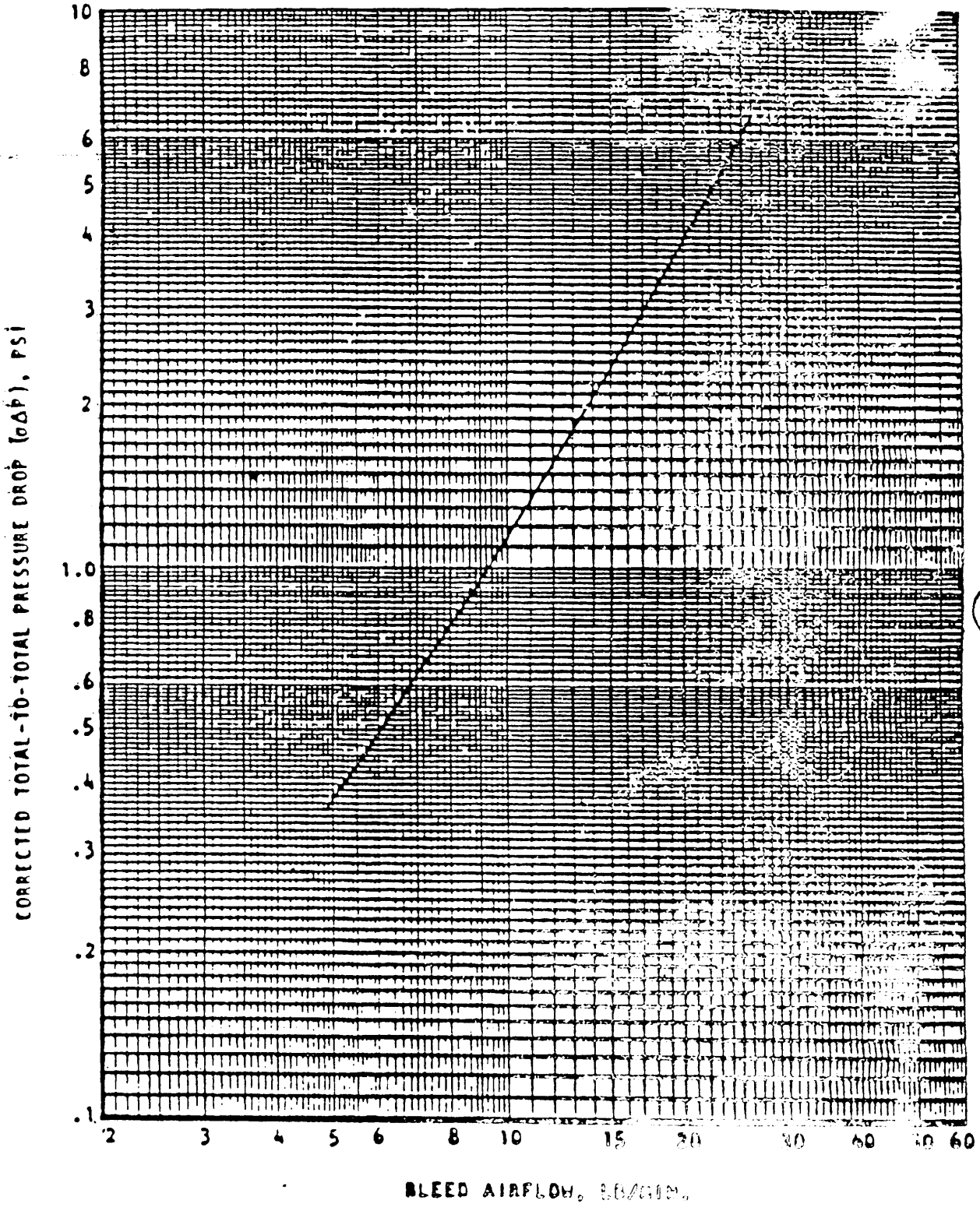


FIGURE 2. Allowable heat exchanger bleed air pressure loss.

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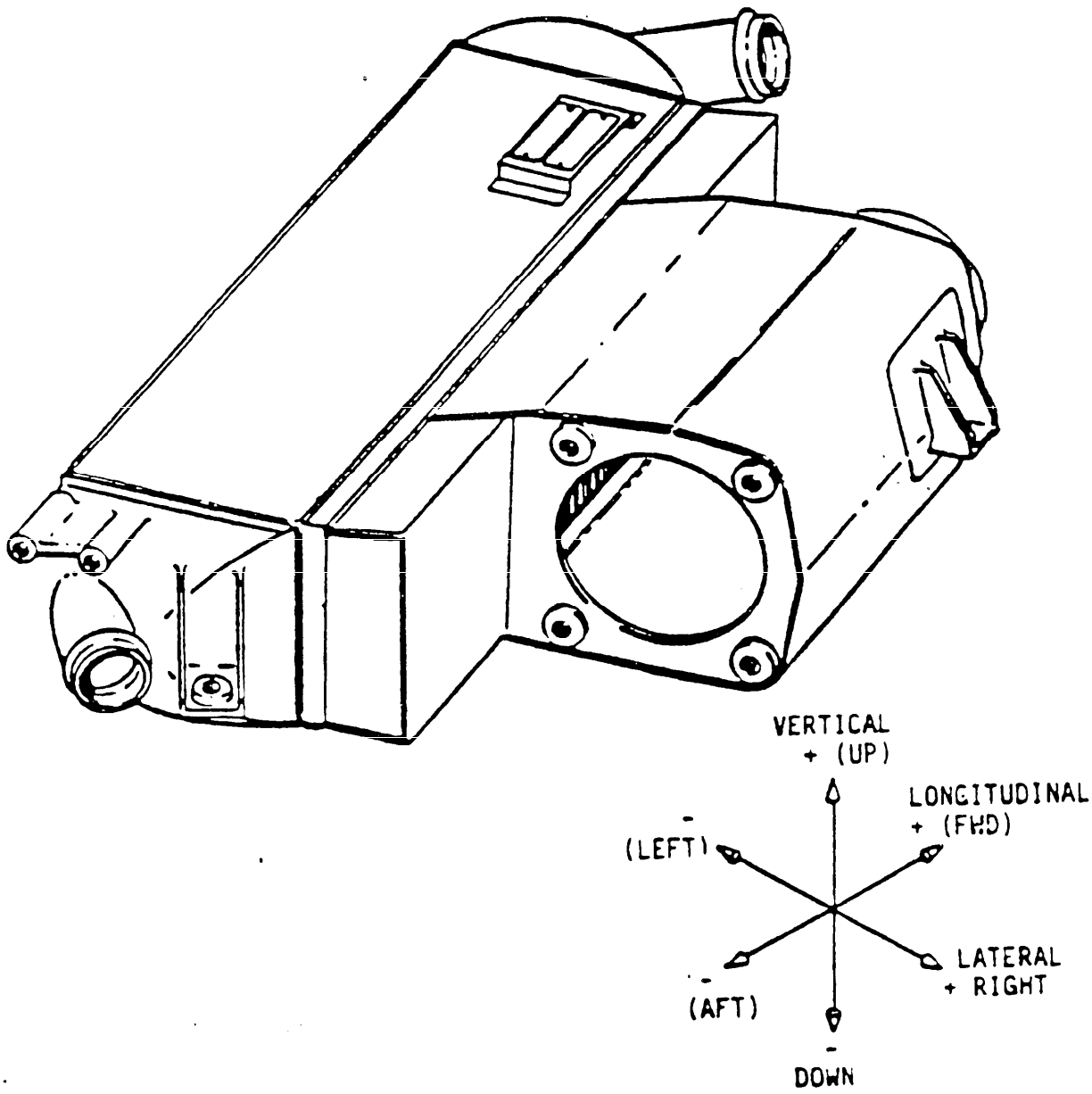


FIGURE 3. Axes orientation.

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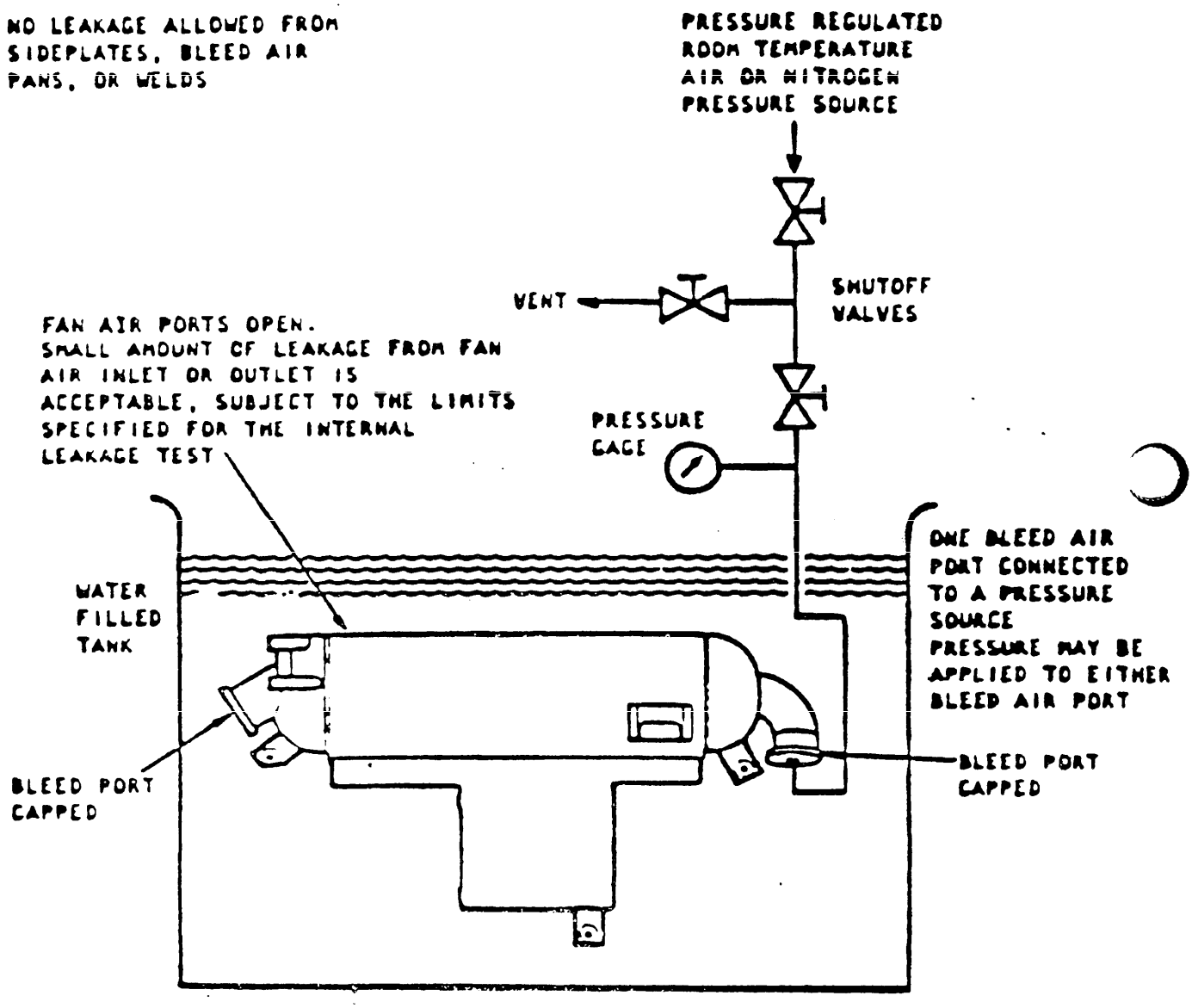


FIGURE 4. Leakage test setup.

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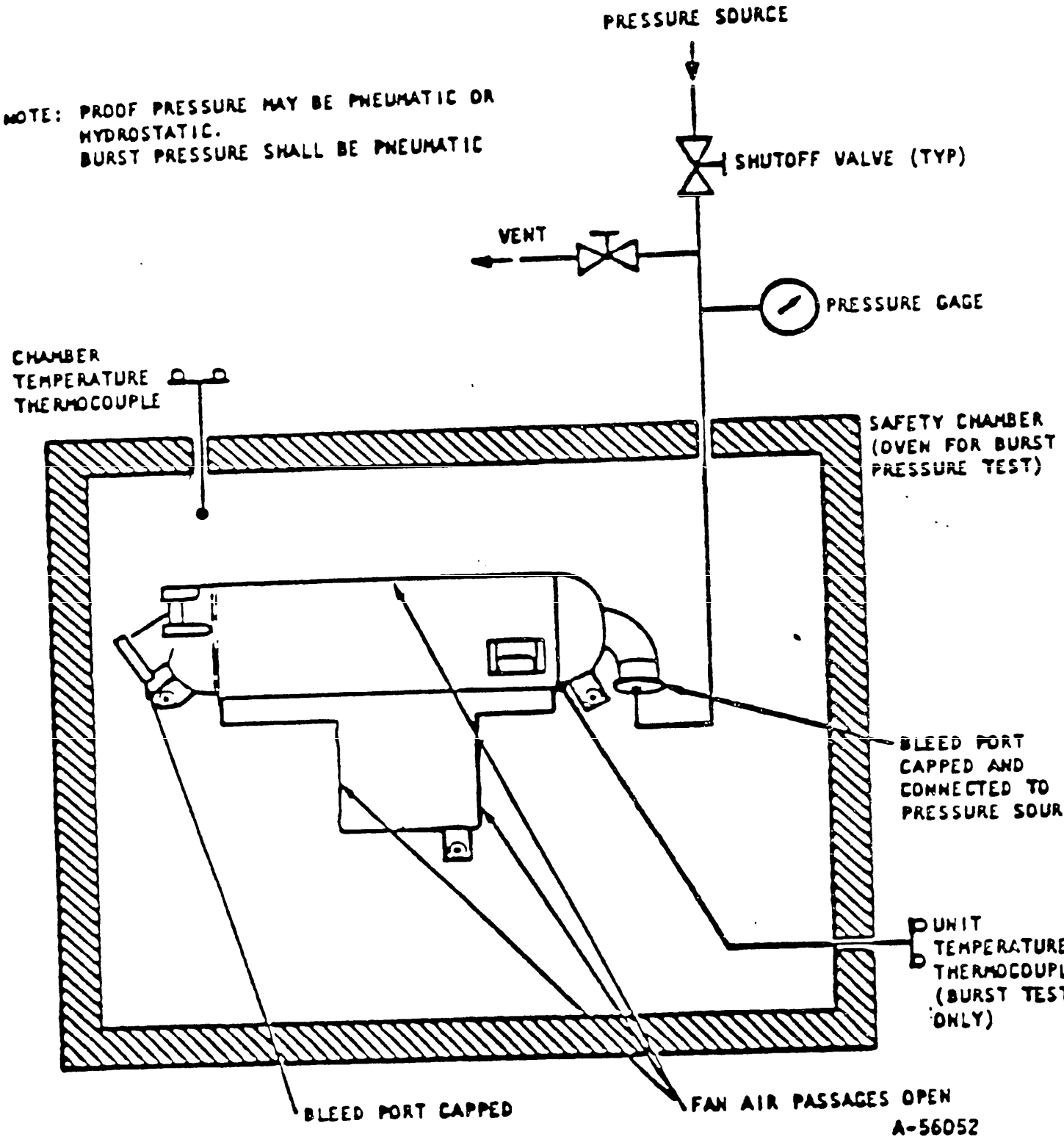


FIGURE 5. Proof and burst pressure test setup.

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